6.3.2 Inventory control

Inventory is any stored resource that is used to satisfy a current or a future need. Raw materials, finished goods are examples of inventory. The inventory problem involves placing and receiving orders of given sizes periodically so that the **total cost of inventory** is minimized.

An inventory decisions

1. How much to order ? 2. When to order ?

Costs involved in an inventory problems

(i) Holding cost (or) storage cost (or) inventory carrying cost (C_1) :

The cost associated with carrying or holding the goods in stock is known as holding cost per unit per unit time.

(ii) Shortage cost (C_2) :

The penalty costs that are incurred as a result of running out of stock are known as shortage cost.

(iii) Setup cost (or) ordering cost (or) procurement cost (C_3) :

This is the cost incurred with the placement of order or with the initial preparation of production facility such as resetting the equipment for production.

6.3.3 Economic Order Quantity (EOQ):

Economic order quantity is that size of order which minimizes total annual cost of carrying inventory and the cost of ordering under the assumed conditions of certainty with the annual demands known. Economic order quantity (EOQ) is also called Economic lot size formula.

The derivation of this formula is given for better understanding and is exempted from examination.

The formula is to determine the optimum quantity ordered (or produced) and the optimum interval between successive orders, if the demand is known and uniform with no shortages.

Let us have the following assumptions.

(i) Let R be the uniform demand per unit time.

(ii) Supply or production of items to the inventory is instantaneous.

(iii) Holding cost is ₹ C_1 per unit time.

(iv) Let there be 'n' orders (cycles) per year, each time 'q' units are ordered (produced).

(v) Let ₹ C_3 be the ordering (set up) cost per order (cycle). Let 't' be the time taken between each order.

Diagrammatic representation of this model is given below:

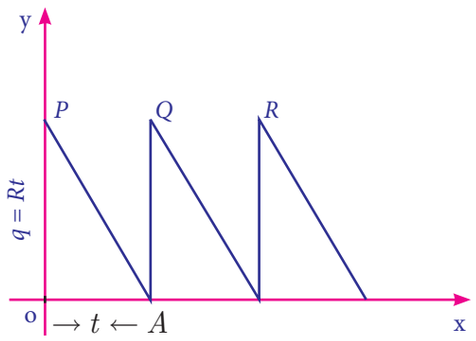


Fig. 6.12

If a production run is made at intervals t, a quantity $q = Rt$ must be produced in each run. Since the stock in small time dt is $Rt\,dt$, the stock in period t is

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